Subtracting Numbers in Scientific Notation

Jonathan R. Bacolod

Scientific Notation

Scientific notation expresses a number in the form

$$a \times 10^n$$

where $1 \leq |a| < 10$ and n is an integer.

Remember: Match, Subtract, Normalize!

Match the exponents.

Subtract the coefficients.

Normalize the result.

Steps (Part 1)

To **subtract numbers in scientific notation**:

1. Check whether the powers of 10 are the same.
2. If the exponents are different, rewrite one or both numbers so that they have the same exponent.

Steps (Part 2)

3. Subtract the coefficients while keeping the common power of 10.
4. Express the final answer in proper scientific notation.

Same Exponents

When the exponents are the same, subtract only the coefficients and retain the common power of 10.

$$(9 \times 10^4) - (2 \times 10^4) = (9 - 2) \times 10^4$$

Different Exponents

When the exponents are different, use the **equivalent forms** of scientific notation to make the exponents equal before subtracting.

$$2.5 \times 10^5 = 0.25 \times 10^6$$

Making Exponents Equal

Exponent	**Decimal Point**
Increases by 1	Moves 1 place left
Decreases by 1	Moves 1 place right

$$2.5 \times 10^5 = 0.25 \times 10^6$$

Normalize the Result

If the resulting coefficient is less than 1 or greater than or equal to 10 in absolute value, adjust the coefficient and exponent to obtain proper scientific notation.

$$0.3 \times 10^5 = 3 \times 10^4$$

Examples

Example 1

Subtract.

$$(9.5 \times 10^4) - (2.3 \times 10^4)$$

Match: Exponents are already the same.

$$= (9.5 - 2.3) \times 10^4$$

Subtract:

$$= 7.2 \times 10^4$$

Example 2

Subtract.

$$(6.4 \times 10^6) - (2.5 \times 10^5)$$

Match:

$$2.5 \times 10^5 = 0.25 \times 10^6$$

Example 2 (continued)

Subtract:

$$(6.4 - 0.25) \times 10^6$$

$$= 6.15 \times 10^6$$

Normalize: Already in proper scientific notation.

Example 3

Subtract.

$$(3.2 \times 10^5) - (2.9 \times 10^5)$$

Subtract:

$$= 0.3 \times 10^5$$

Normalize:

$$= 3 \times 10^4$$

Example 4

Subtract.

$$(5.1 \times 10^{-3}) - (4.2 \times 10^{-4})$$

Match:

$$4.2 \times 10^{-4} = 0.42 \times 10^{-3}$$

Example 4 (continued)

Subtract:

$$(5.1 - 0.42) \times 10^{-3}$$

$$= 4.68 \times 10^{-3}$$

Normalize: Already in proper scientific notation.

Practice Exercises

Practice 1

Subtract the following numbers expressed in scientific notation.

$$(8.7 \times 10^5) - (3.4 \times 10^5)$$

Practice 2

Subtract the following numbers expressed in scientific notation.

$$(9.6 \times 10^7) - (2.1 \times 10^7)$$

Practice 3

Subtract the following numbers expressed in scientific notation.

$$(7.5 \times 10^6) - (4.2 \times 10^5)$$

Practice 4

Subtract the following numbers expressed in scientific notation.

$$(6.8 \times 10^8) - (3.5 \times 10^7)$$

Practice 5

Subtract the following numbers expressed in scientific notation.

$$(5.4 \times 10^4) - (2.7 \times 10^3)$$

Practice 6

Subtract the following numbers expressed in scientific notation.

$$(4.2 \times 10^9) - (1.6 \times 10^8)$$

Practice 7

Subtract the following numbers expressed in scientific notation.

$$(3.6 \times 10^{-3}) - (1.4 \times 10^{-3})$$

Practice 8

Subtract the following numbers expressed in scientific notation.

$$(8.5 \times 10^{-4}) - (2.3 \times 10^{-5})$$

Practice 9

Subtract the following numbers expressed in scientific notation.

$$(6.7 \times 10^{-6}) - (3.2 \times 10^{-7})$$

Practice 10

Subtract the following numbers expressed in scientific notation.

$$(9.2 \times 10^{-2}) - (4.8 \times 10^{-3})$$

Activities

Activity 1

Subtract the following numbers expressed in scientific notation.

$$(7.9 \times 10^5) - (2.6 \times 10^5)$$

Activity 2

Subtract the following numbers expressed in scientific notation.

$$(8.4 \times 10^7) - (3.2 \times 10^7)$$

Activity 3

Subtract the following numbers expressed in scientific notation.

$$(6.8 \times 10^6) - (3.1 \times 10^5)$$

Activity 4

Subtract the following numbers expressed in scientific notation.

$$(9.3 \times 10^8) - (4.6 \times 10^7)$$

Activity 5

Subtract the following numbers expressed in scientific notation.

$$(7.2 \times 10^4) - (3.5 \times 10^3)$$

Activity 6

Subtract the following numbers expressed in scientific notation.

$$(5.7 \times 10^9) - (2.4 \times 10^8)$$

Activity 7

Subtract the following numbers expressed in scientific notation.

$$(4.8 \times 10^{-3}) - (2.1 \times 10^{-3})$$

Activity 8

Subtract the following numbers expressed in scientific notation.

$$(7.6 \times 10^{-4}) - (3.2 \times 10^{-5})$$

Activity 9

Subtract the following numbers expressed in scientific notation.

$$(8.3 \times 10^{-6}) - (2.5 \times 10^{-7})$$

Activity 10

Subtract the following numbers expressed in scientific notation.

$$(6.4 \times 10^{-2}) - (2.7 \times 10^{-3})$$

Thank you!